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B.Tech. Degree VI Semester Special Supplementary Examination November 2022

ME 19-205-0601 DYNAMICS OF MACHINERY
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Recall the concepts of free body diagrams, principles of statics and dynamics.
 CO2: Use graphical and analytic methods to do force analysis of planar mechanisms.
 CO3: Apply these concepts in different machine elements for the evaluation of forces and moments.
 CO4: Analyze the dynamics of different mechanisms and machine elements and determine the various forces and torques.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

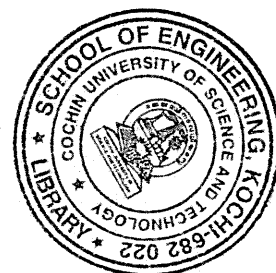
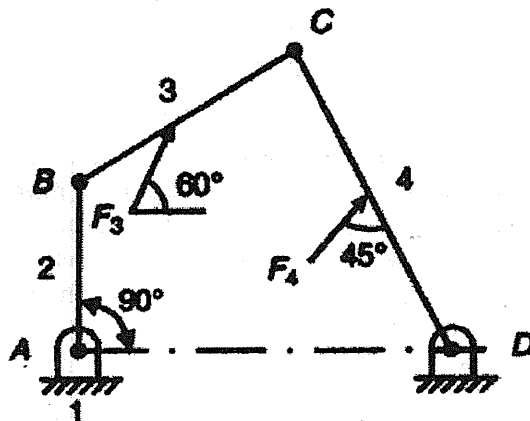
(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) State and explain D'Alembert's principle.	3	L2	1	1
(b) What is meant by piston and crank effort?	3	L1	2	1,2
(c) Define coefficient of fluctuation of energy and coefficient fluctuation of speed.	3	L1	3	1,2,3
(d) Explain the effect of gyroscopic couple on a naval ship.	3	L2	3	1,2,3
(e) What do you mean by primary and secondary unbalance in reciprocating engines?	3	L1	3	1,2,3
(f) Define hammer blow, tractive effort and swaying couple.	3	L2	3	1,2,3
(g) Explain the phenomena of 'slip' and 'creep' in a belt drive.	3	L2	3	1,2,3
(h) Explain the meaning of sensitiveness, hunting and stability of a governor.	3	L2	3	1,2,3

PART B

(4 × 12 = 48)

- II. (a) A four bar mechanism as shown in figure is subjected to two forces, $F_3 = 1000$ N at 60° from horizontal and $F_4 = 2000$ N at 45° from link 4. The dimensions of links are as under:
 $AB = 300$ mm, $BC = 400$ mm, $CD = 450$ mm and $AD = 600$ mm
 Perform static force analysis and determine resisting torque on link 2.



- (b) Define the terms inertia force and inertia couple. How is their magnitude and direction found?

OR

(P.T.O.)

		Marks	BL	CO	PO
III.	(a) A vertical cylinder petrol engine has a bore of 100 mm and stroke of 120 mm. The length of connecting rod between centres is 250 mm. The mass of the piston is 1.1 kg. The speed of engine is 1500 rpm. In the expansion stroke with a crank at 30° from TDC, the gas pressure is 700 kN/m^2 . Determine: (i) net force on the piston (ii) force on the connecting rod (iii) thrust on the cylinder wall (iv) crank effort (v) speed above which the gudgeon pin force would reverse in direction.	9	L4	4	1,2,3
	(b) What is dynamically equivalent system?	3	L1	1	1
IV.	(a) Write the procedure for determining the turning moment diagram.	5	L1	3	1,2,3
	(b) A double-acting steam engine develops 56 kW of power at 210 rpm. The maximum and minimum speeds do not vary more than 1% of the mean speed and the excess energy is 30% of the indicated work per stroke. Determine the mass of the flywheel if the radius of gyration of the flywheel is 500 mm.	7	L4	3	1,2,3
OR					
V.	(a) The rotor of a marine turbine has a moment of inertia of 750 kg.m^2 and rotates at 3000 rpm clockwise when viewed from aft. If the ship pitches with angular simple harmonic motion having a periodic time of 16 seconds and an amplitude of 0.1 radian, find: (i) the maximum angular velocity of the rotor axis (ii) the maximum value of the gyroscopic couple (iii) the gyroscopic effect as the bow dips	7	L3	3	1,2,3
	(b) How a four-wheeled vehicle is affected by gyroscopic couple?	5	L1	3	1,2,3
VI.	Four masses 150, 250, 200 and 300 kg are rotating in the same plane at radii of 0.25, 0.2, 0.3 and 0.35 m respectively. Their angular location is 40° , 120° and 250° from mass 150 kg, respectively, measured in counter-clockwise direction. Find the position and magnitude of the balance mass required, if its radius of rotation is 0.25 m.	12	L3	4	1,2,3
OR					
VII.	A single cylinder reciprocating engine has speed 240 rpm, stroke 300 mm, mass of reciprocating parts 5 kg and mass of revolving parts at 150 mm radius is 3.7 kg. If two-third of the reciprocating parts and all the revolving parts are to be balanced, find: (i) the balancing mass required at a radius of 400 mm (ii) the residual unbalanced force when the crank has rotated 60° from top dead centre.	12	L3	4	1,2,3
VIII.	Two parallel shafts that are 3.5 m apart are connected by two pulleys of 1-m and 400-mm diameters, the larger pulley being the driver runs at 220 rpm. The belt weighs 1.2 kg per metre length. The maximum tension in the belt is not to exceed 1.8 kN. The coefficient of friction is 0.28. Owing to slip on one of the pulleys, the velocity of the driven shaft is 520 rpm only. Determine: (i) the torque on each shaft (ii) the power transmitted (iii) the power lost in friction (iv) the efficiency of the drive.	12	L3	4	1,2,3

OR

(Continued)

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		Marks	BL	CO	PO
IX.	(a) What is the main function of a governor? How does it differ from that of a flywheel?	4	L2	4	1,2,3
	(b) Each arm of a Porter governor is 200 mm long and is pivoted on the axis of the governor. The radii of rotation of the balls at the minimum and the maximum speeds are 120 mm and 160 mm respectively. The mass of the sleeve is 24 kg and each ball is 4 kg. Find the range of speed of the governor. Also determine the range of speed if the friction at the sleeve is 18 N.	8	L3	4	1,2,3

Bloom's Taxonomy Levels

L1 = 20.8%, L2 = 15.8%, L3 = 42.5%, L4 = 20.8%.
